

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. An apparent gain is uninformative if the answer can be read from the primer without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate primers against four controls jointly: a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring that varies triangle count at fixed prompt length. Across four Qwen3 arms (1.7B and 4B, with and without thinking mode) on 40-node Erdős–Rényi graphs, primers that state the answer help a model at floor and harm a model at ceiling (cross-fitted interaction with baseline accuracy -11.0 points, 95% CI $[-17.1, -5.7]$), whereas primers that do not state it show no such relation. The pattern replicates on eleven screened arms (0.6–14B, two families) but is not graded at intermediate baselines. The one primer that does not state the answer and still beats both controls, `clustering on node_degree`, gains $+5.9$ points at intermediate densities and little elsewhere; two candidate mechanisms for it are not supported.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because

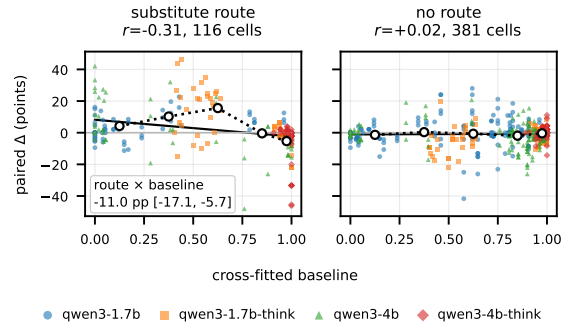


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.3) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right), each shown once per fold direction. Baseline and effect come from disjoint halves of the paired graphs, so shared sampling noise cannot produce the slope.

the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task leaves no room for any primer effect to appear.

We address three research questions. **RQ1:** when a primer improves accuracy, is the improvement attributable to the primer stating the answer? **RQ2:** does added text affect accuracy independently of its content? **RQ3:** does any primer that does not state the answer improve accuracy beyond a length-matched control, and through what mechanism? Our main contribution is the four controls of Figure 2 applied jointly — a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched `filler`, a minimal-length `components`, and a degree-preserving rewiring (Section 3.3) — on 40-node Erdős–Rényi

graphs across four model arms and six tasks.

2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 2; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so they share sentence framing and number formatting and differ only in the facts stated. The exception is *filler*, which states no fact and therefore uses its own frame (“Node N is simply present within the graph G .”); it is a length control, not a content control. *rwse* states each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)).

3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that

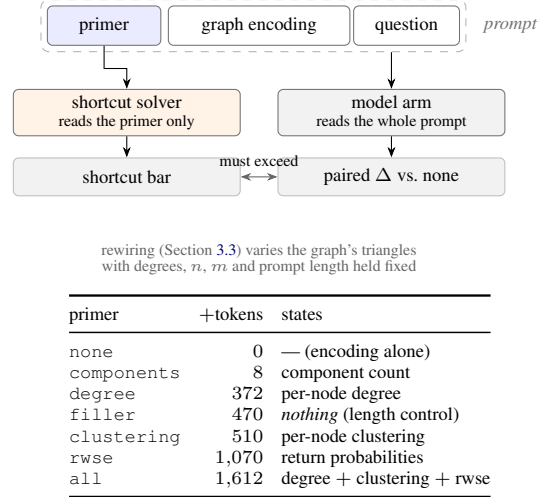


Figure 2: Experimental design. One prompt, two readers: the graph-blind shortcut solver sees only the primer and sets the bar a primer’s gain must exceed; the model arm sees all of it. Lengths are means at $n=40$ (Qwen3 tokenizer) against a mean *none* prompt of 2,165 tokens, so *filler* is a close length control for *clustering* but bounds only the length term for *rwse* and *all*.

never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Its accuracy on a (task, condition) cell, the *shortcut bar*, is a content-only lower bound: it never overstates what the primer text alone permits.

Bars are fit at the evaluated graph size: $n=40$ Erdős–Rényi graphs at the seven densities of the sweep, with disjoint 300-graph fit and test sets per density (Table 1); the screened arms, which answer the published 5–19-node split, use bars fit on that split, which differ substantially (Appendix Table 8). A (task, condition) pair offers a *substitute route* when its bar exceeds the *none* bar by more than 0.05; every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Only *node_degree* (under *rwse*, *degree*, *all*) and *edge_count* (under *clustering*, *degree*, *all*) offer a route. *node_count* and *cycle_check* have constant gold answers at $n=40$ (40, and “yes” at every density) and are excluded.

3.3 Rewiring: isolating structure from length

filler controls length only approximately, and comparisons across graph generators confound structure with class balance, so we use a within-

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 1: Shortcut bars at $n=40$ (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

instance design. A double-edge swap replaces edges $(a, b), (c, d)$ with $(a, d), (c, b)$ (Fosdick et al., 2018), which preserves the degree sequence, n, m , the node_degree gold answer and the prompt length while changing the triangle count that the clustering primer reports. We generate low, base and high variants of each graph ($n=40$, mean degree 12, 200 graphs per level; mean clustering 0.01, 0.31, 0.73) and evaluate node_degree under none, clustering and filler with qwen3-1.7b, qwen3-1.7b-think and qwen35-2b. If the clustering benefit derives from the stated values, it should vary with triangle count. If instead the model reads mean clustering as a density prior (in an Erdős-Rényi graph $\bar{C} \approx p$), its degree estimates should shift toward $\bar{C}(n-1)$: 0.5, 12.0 and 28.7 across levels, against a true mean of 12.

3.4 Answer extraction and measurement validity

Extraction errors can masquerade as condition effects, so answers are extracted with task-specific patterns rather than a positional rule: on cycle_check a last-yes/no rule is not sufficient, because qwen3-4b often answers correctly and then recites that a cycle has “no repeated” nodes, which a positional rule scores against one condition more than another. A manual check of 50 rows per arm confirmed that <think> blocks never reach the extractor. Truncated pairs are dropped; scoring them as incorrect changes the verdict on only three qwen3-1.7b cycle_check cells (Appendix A).

3.5 Significance testing

Every comparison is paired on the same graph and question (Dror et al., 2018) and tested with the exact McNemar test (McNemar, 1947), Benjamini-Hochberg corrected (Benjamini and Hochberg,

1995) within each (arm, task) family at $q=0.05$; a claim that a condition beats both controls also requires significance under a global correction. For the effect-baseline relation (Section 5.3), baseline and effect are estimated from disjoint random halves of each cell’s graphs (Chernozhukov et al., 2018), and the route $\times$ baseline interaction is bootstrapped over (arm, task, density) blocks, which share one none sample. For null cells we simulate a minimum detectable effect (MDE) at power 0.8, $\alpha=0.05$.

4 Experimental Setup

Tasks and arms. The six GraphQA tasks (Fatemi et al., 2023) are those named in Table 2. An arm is a model checkpoint with a fixed decoding configuration, so native thinking mode off and on gives two arms of one checkpoint.

Starting point. We begin where the encoding question was posed, on the published GraphQA split (Fatemi et al., 2023): 5–19 nodes, 30 graphs per cell, five tasks. Eight arms from two families answer it — Qwen3-8B and Qwen3-14B (Qwen Team, 2025), Gemma-4-E4B-it and Gemma-4-12B-it (Google DeepMind, 2026), each with thinking off and on. The corpus cannot carry the question. Under none every arm already scores 0.97–1.00 on four of the five tasks, and only edge_count separates the arms at all (Appendix Table 5), so a primer has almost nothing left to add and a null result would be uninformative. The experiment therefore moves to graphs large and dense enough to leave headroom in both directions.

Main sweep. Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025) carry it, each with thinking off (qwen3-1.7b, qwen3-4b) and on (-think): two sizes, both modes of the same weights, and the two sizes that retain headroom on five of six tasks at $n=40$. All four arms answer one shared prompt set of Erdős-Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at densities $p \in \{.10, .20, .35, .50\}$, 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm, extended on node_degree and edge_existence to $p \in \{.65, .75, .85\}$ (100 graphs per cell, all four arms) so that the density range reaches from near-ceiling to near-floor. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.5 reports a budget-matched comparison. Every headline result below

is measured on this sweep.

Supporting corpora. Four further corpora each answer one question the main sweep cannot, and each is used only for that question. (i) A dedicated `node_degree` sweep with `qwen3-1.7b` and `qwen3-1.7b-think` — 400 graphs per density at all seven densities, with a `filler` arm, plus an independent replication on 400 newly seeded graphs per density at $p \leq .50$ — supplies the power the main sweep lacks for the one non-answer content effect (Section 5.4). (ii) A degree-preserving rewiring corpus (Section 3.3; `qwen3-1.7b`, `qwen3-1.7b-think` and `Qwen3.5-2B` (Qwen Team, 2026), `qwen3.5-2b`) tests whether that effect tracks the triangle count the primer reports. (iii) A fixed-mean-degree grid — mean degree ≈ 8 or ≈ 16 with $n \in \{20, 40, 80, 160\}$ on `node_degree`, answered by `qwen3-1.7b` and `qwen3-8b` — separates answer size from graph size and tests the same effect across four graph sizes (Section 5.7); it carries no `filler` arm. (iv) A 100-graph probe of `none` and `degree` on all six tasks, answered by `Qwen3.0.6B` with and without thinking and by `qwen3.5-2b`, adds a third family and a smaller size to the robustness estimate in Appendix Table 13.

Screened arms. The eleven arms of the starting point and the probe support the two checks of Section 5.3; their shortcut bars are refit on their own corpus (Appendix Table 8).

Metrics. Exact match for every task; mean absolute error for the integer tasks is in the Appendix.¹

5 Results

5.1 Main sweep

Table 2 gives the main sweep, and its largest entries are not primer effects. They fall on `node_count`, whose gold answer is the same for every graph, and on the two truncated `edge_count` cells, both flagged in the table. Across the four remaining tasks the entries are small and often change sign between the two model sizes. Two systematic patterns account for most of them: a cost of added text, which `filler` isolates (Section 5.2), and a benefit that appears only where the primer states the answer (Section 5.3). Only one arm is out of range: `qwen3-4b-think` scores 0.976–1.000 under

`none` on all six tasks, so no primer can move it. The other three retain headroom, and thinking does not by itself remove it — `qwen3-1.7b-think` is at 0.642 on `node_degree` and 0.797 on `connected_nodes`, which is where its content effect is measured (Section 5.4).

Throughout, Δ is a paired difference against `none` in percentage points, computed over instances in which neither condition was truncated; per-cell bootstrap intervals are in the Appendix. Outside the two flagged cells, every task truncates on fewer than 7% of instances.

5.2 Length is an active treatment

`filler` significantly reduces accuracy on `connected_nodes` and `edge_existence` in both plain arms, even though it states no fact about the graph. The cost is not a flat charge for tokens. It differs by task by a factor of two (Section 5.7, Appendix Table 7a), and within a task it is not monotone in density: on `qwen3-1.7b/edge_existence` it runs $-5, -20, -32, -6, -2, -1, -1$ from $p=.10$ to $.85$, largest in the middle of the range and gone where the task collapses toward a constant answer. Whether it tracks difficulty depends on the arm: correlating the cost with accuracy under `none` gives $+0.55$ for `qwen3-1.7b-think`, -0.09 for `qwen3-1.7b` and -0.44 for `qwen3-4b`, so the relation the dedicated `node_degree` sweep shows on the thinking arm ($r=+0.95$ over its seven densities; 6.3 points pooled, $p<10^{-4}$) is absent on one plain arm and reversed on the other.

Effect size is better predicted by the headroom a cell leaves than by its density. Over the 126 uncontaminated non-degenerate cells of the two plain arms, $|\text{content gain}|$ correlates $+0.46$ $[0.28, 0.60]$ with headroom ($\min(a, 1-a)$) on `none`, read off the control alone) holding density fixed, against -0.21 $[-0.37, -0.02]$ for density holding headroom fixed. Density is thus not only a proxy, but it is the weaker of the two. Within `edge_existence`, where the prior is U-shaped in density and the two are uncorrelated ($r=0.04$), the point estimates agree but 14 blocks cannot separate them.

The `edge_existence` effects are not changes in how the graph is read. The hit rate stays at 99–100% under every condition, so accuracy moves only through the false-alarm rate, which `filler` raises and `clustering` lowers. With hits at ceiling, sensitivity and criterion cannot

¹Set-F1 on `connected_nodes` is 0.96–1.00 in every cell and does not discriminate between conditions.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check [†]	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count [†]	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	+0.0	-3.0	-12.5	-2.8
	cycle_check [†]	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count [†]	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-8.0
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	-0.3	+7.4	+2.0	+6.1
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	-0.7	+0.0	+0.0
	edge_count [‡]	86.3	+7.4	-8.7	-4.0	-61.5	+3.4	-8.3
	edge_existence	99.2	-0.8	-1.0	-1.3	-2.3	-0.8	-2.0
	node_count [†]	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	-1.0	+1.0	+2.5	+11.7	-3.5	+11.6
qwen3-4b-think	connected_nodes	98.7	+0.5	+0.0	+0.0	+0.0	+0.8	-0.5
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	-0.8	+0.0	+0.0
	edge_count [‡]	97.6	+1.7	+1.7	+0.0	-23.5	-2.0	-41.5
	edge_existence	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_count [†]	99.5	+0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	-0.3	-0.3	-2.1	+0.0	+0.0	-0.8

Table 2: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points, truncated generations excluded pairwise. [†]: gold answer is constant at $n=40$, so a shift need not reflect graph reading (Section 5.6). [‡]: $\geq 62\%$ of generations reach the output budget (Table 9). qwen3-4b-think is at ceiling on every task, leaving no headroom for a primer effect; the other three arms are not (Section 5.1). $n \in [13, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini-Hochberg within each (arm, task) family, $q = 0.05$.

be separated (Macmillan and Creelman, 2005; Hautus, 1995) (Appendix Tables 10 and 11). The node_count gains are not a primer effect either (Section 5.6).

5.3 Answer-stating primers: route substitution

A cell is an (arm, task, density, condition) combination with at least ten usable pairs. On the main sweep plus the high-density extension, route cells gain where accuracy under none is low and lose where it is high, while no-route cells show no such relation (Figure 1; binned means also in Appendix Table 12). Baseline and effect share sampling noise, and a difference in significance is not a significant difference (Gelman and Stern, 2006), so we estimate the route $\times$ baseline interaction on cross-fitted halves of each cell’s graphs: -11.0 points per unit of baseline, with logit-scale and windowed robustness in Appendix Table 13. The relation is also not an artifact of censoring. It pools two tasks, one of which (edge_count) truncates on 22–81% of generations, so we partition it. Counting each route

cell once per fold direction, as Figure 1 does, gives 232 points: over all of them $r = -0.308$ with slope -10.5 ; over the 168 from node_degree alone $r = -0.235$ ($p=0.0022$) with slope -9.1 ; over the 64 from edge_count $r = -0.664$ with slope -26.5 . node_degree truncates on 0–1.6% and carries the relation on its own. The relation is not graded. Restricted to cells whose baseline lies between 0.10 and 0.90 it disappears altogether, so what the data support is a contrast between floor and ceiling rather than a smooth dependence on competence: a stated answer helps a model that cannot compute the answer and hurts one that already can. Our coverage is thinnest exactly where a graded relation would have to show itself. Manipulating difficulty directly reproduces the contrast. For qwen3-1.7b-think on node_degree, raising density from $p=.10$ to .85 drives accuracy under none down and degree’s benefit up almost monotonically, from $+3.1$ to $+34.2$ points (Figure 3a).

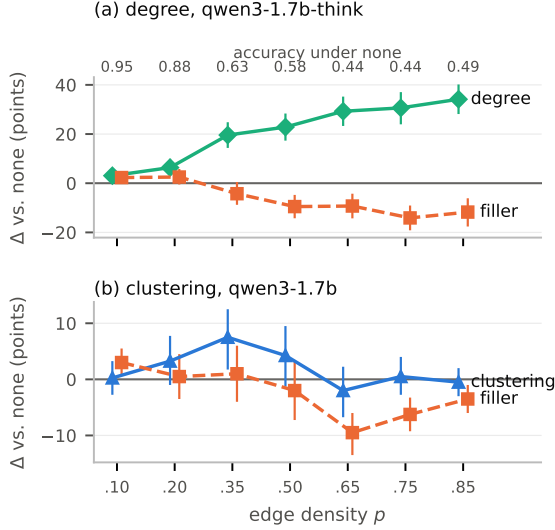


Figure 3: Paired effect against `none` on `node_degree` by edge density (400 graphs per density, 95% bootstrap intervals), on a shared density axis. (a) the answer-stating `degree` primer climbs as the task gets harder; (b) the non-answer `clustering` primer peaks at intermediate density. Note the $4\times$ difference in y scale between the panels: the content effect is an order of magnitude smaller than the leakage effect. `filler` is in both panels as the length control.

Screened arms. Two checks on the eleven screened arms point the same way. The interaction has the same sign and direction there, though it is estimated far less precisely (Appendix Table 13). Substituting the other arms’ accuracy on a cell for the cell’s own baseline removes the relation, so it tracks a model’s own competence rather than the difficulty of the item. Neither separates copying a stated value from computing the answer, but one sign of copying is that when a model errs despite a stated answer, the value it gives is its list-neighbour’s stated degree far more often than chance allows ($p=.002$).

5.4 Non-answer content: clustering on `node_degree`

Two primers that do not state the answer exceed both controls; the second is `components` on `connected_nodes` (Section 5.7). That count pools densities; repeating the scan per density under the same global correction leaves six non-route cells, four of them on `edge_existence`, where a gain is a criterion shift rather than better graph reading (Section 5.2), and two that are these same two effects at a single density. `clustering` on

`node_degree` is the one with a dedicated replication. This cell and the $p=.35-.50$ window below were chosen after the main sweep, so the cell was re-run with 400 graphs per density and an independent-seed replication that played no part in either choice (Figure 3b; Appendix Tables 14 and 15). Pooled over all seven densities the gain over `none` is small (+1.9 points), and the apparently larger gain over `filler` mostly reflects `filler`’s own cost, which accounts for the whole difference at the densest levels. The effect is concentrated in the middle of the density range. Pooled over $p=.35$ and $.50$, `clustering` exceeds `none` by +5.9 points (95% CI [2.1, 9.8]) and `filler` by a similar margin, while `filler` and `none` do not differ there, so length does not account for it. The replication gives +5.5 over the same two densities, and per-density estimates are too imprecise to localise the effect more finely. `qwen3-4b` is at ceiling in the main sweep but shows the effect on the high-density extension, under the global correction against both controls. On the thinking arms the effect is weak rather than absent: +2.3 points at $p \leq .50$ ($p=0.036$), and +5.0 against the length control.

Neither mechanism we can test explains the effect. Under rewiring, mean clustering rises from 0.01 to 0.73 while the benefit stays flat, and for `qwen35-2b` `filler` matches or exceeds `clustering` at every level (Appendix Figure 4). The density-prior account fails as well. Signed error under `clustering` should shift by up to 17 at the high level, but the paired shift stays within 0.3 of zero at every level for all three arms (Appendix Table 16).

5.5 Thinking mode buys accuracy with budget

Thinking brings most cells to ceiling, and where it does not, truncation censors the task. Part of its advantage is the larger output budget rather than the reasoning it buys. Rescored at the plain arms’ 2,048-token budget, which is exact under greedy decoding, the 1.7B thinking arm still wins on `edge_existence` (97.8 against 70.5%) but loses on `cycle_check` (38.0 against 90.7%; Appendix Table 17).

Many of the main-sweep nulls are uninformative: `none` is already at floor or ceiling in 42% of `qwen3-1.7b`’s null cells and in all 33 of `qwen3-4b-think`’s. Where a gain could be detected, the median detectable size is 5.6 points for the 1.7B arms and 10.0 for 4B.

n	$\bar{d}$	qwen3-1.7b			qwen3-8b	
		none	clust.	Δ	none	Δ
20	8	0.912	0.865	-4.8*	0.998	+0.2
40	8	0.728	0.770	+4.2	0.995	+0.0
80	8	0.645	0.750	+10.5*	0.965	-0.8
160	8	0.624	0.693	+6.9*	0.897	-2.8
20	16	0.575	0.740	+16.5*	0.985	-0.2
40	16	0.398	0.476	+7.8*	0.975	+1.0
80	16	0.225	0.323	+9.8*	0.895	-1.8
160	16	0.224	0.211	-1.3	0.802	-4.0
pooled		0.542	0.604	+6.2	$n=3191, p<10^{-4}$	

Table 3: Fixed-mean-degree sweep on `node_degree`: mean degree $\bar{d}$ is held while n varies, so density falls as the prompt lengthens. Growing n eightfold at $\bar{d}\approx 8$ costs 29 points and saturates after $n=80$; doubling $\bar{d}$ costs 33–42 points at every n . Δ is `clustering` against `none`, exact match. *: Benjamini–Hochberg at $q=0.05$ across the eight cells. This corpus has no `filler` arm, so Δ is a net effect. qwen3-8b is a ceiling control.

5.6 What makes these tasks hard

A primer can only move accuracy where accuracy can move. Four things set that room at $n=40$, and none of them is the size of the graph.

The size of the answer, not the size of the graph. A sweep that holds mean degree at ≈ 8 or ≈ 16 while n ranges over $\{20, 40, 80, 160\}$ separates two factors that a density sweep confounds (Table 3). Growing n eightfold at $\bar{d}\approx 8$ costs qwen3-1.7b 29 points and saturates after $n=80$; doubling $\bar{d}$ costs 33–42 points at *every* n . The same grid puts qwen3-8b at 0.998–0.802 throughout, so scale removes the headroom before graph size exhausts it.

The output budget, on tasks that require enumeration. On `edge_count` under `none` the plain arms almost never truncate and almost never answer exactly — qwen3-4b is exact on 8 of 400 with a median relative error of 10.9%, qwen3-1.7b exact on 5 with 28.5% — while the thinking arms are exact on 86% and 98% of the generations that terminate and truncate on 82% and 80% of all of them. The task is not beyond the plain arms’ competence; without an explicit serial count they substitute an estimate, and with one they run out of budget.

The label distribution, which two tasks do not have. At $n=40$ every one of the 400 Erdős–Rényi graphs contains a cycle and every one has 40 nodes, so `cycle_check` and `node_count`

Cond.	vs none		vs filler	
	Δ	p	Δ	p
components	+9.2	10^{-4}	+21.8	10^{-4}
clustering	+1.0	0.689	+13.5	10^{-4}
rwse	+0.0	1.000	+12.5	10^{-4}
degree	-3.0	0.188	+9.5	0.000
all	-2.8	0.251	+9.8	10^{-4}
filler	-12.5	10^{-4}	—	—

Table 4: `connected_nodes`, qwen3-4b, main sweep pooled ($n=400$), exact match, points. Every condition beats `filler`, because `filler` itself costs 12.5 points; only `components` also beats `none`, and it adds 8 tokens where `filler` adds 470 (Figure 2). **Bold**: survives the global correction against both controls.

both have a majority-class baseline of 1.000: a constant answer is a perfect answer. `cycle_check` is consequently uninformative here (all four arms score 0.997–1.000). `node_count` is worse than uninformative: qwen3-1.7b scores 0.315 against that baseline of 1.000, because 68.5% of its answers read the encoder’s “nodes 0, . . . , 39” preamble as the count. Almost any interposed text interrupts that off-by-one, which is what the large primer gains on this task measure.

The metric, which can hide the ceiling. Set-F1, the metric `connected_nodes` is usually reported on, awards partial credit on a large answer set and sits at 0.96–0.99 at every density. Exact match does not: qwen3-1.7b falls from 0.920 to 0.480 over the same range (Appendix Table 6). Reported on F1 the task looks solved and no primer effect could appear; reported on exact match it has 44 points of headroom. This is why we report exact match throughout (Section 4).

5.7 What content buys, and what it costs

clustering holds across graph sizes. On the fixed-degree grid of Table 3, `clustering` gains +6.2 points over `none` ($n=3,191$; 504 helped against 306 hurt; $p<10^{-4}$), surviving correction in six of eight cells, and is larger where the answer is larger (+8.2 at $\bar{d}\approx 16$ against +4.2 at ≈ 8). The grid played no part in selecting the condition, which the $p=.35-.50$ window of Section 5.4 did. It carries no `filler` arm, so this is a net effect and not evidence that content outweighs length.

The second surviving content effect costs eight tokens. `components` on `connected_nodes` gains +9.2 points over

none and +21.8 over filler for qwen3-4b ($n=400$, $p \approx 10^{-4}$, both under the global correction; Table 4). Every condition beats filler there, because filler itself costs 12.5 points; what distinguishes components is that it also beats none. Since it adds 8 tokens where filler adds 470 (Figure 2), it is the only effect we measure that is substantially free of length cost.

The cost of length does not transfer across tasks, and no ordering of primers by length survives disaggregation. The length cost runs from -9.1 points on edge_existence to -4.3 on node_degree, and changes sign *within* every task (Appendix Table 7a), so it cannot be estimated on one task and applied to another, and there is no per-token rate to quote. The one positive row, node_count, rests on a single cell of a constant-gold task and carries no weight.

Content gain is unstable in the same way (Appendix Table 7b). clustering is worth +10.9 points on qwen3-1.7b, +2.6 on qwen3-1.7b-think and +11.6 on qwen3-4b; rwse runs +11.3, +5.6, +5.2 over the same three arms. A content gain measured on one arm therefore does not carry to another, which is a limit on how far any single-model primer result generalises. Only components holds its value across all three (+5.7, +6.8, +5.5) — the same condition that carries the effect above. On qwen3-4b alone, clustering is worth more than twice rwse; we see that ordering on that arm only and do not find it on the other two.

6 Conclusion

Structural primers change LLM accuracy on GraphQA in two ways. For RQ1, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling; the contrast survives cross-fitting, a logit transformation, screened arms and a negative-control difficulty estimate, but is not graded at intermediate baselines. For RQ2, added text is an active treatment whose cost exceeds most content effects and grows with difficulty, so a primer compared only against none confounds content with length. For RQ3, two effects of non-answer content survive both controls; both are small, one is concentrated at intermediate densities and unexplained. All claims are behavioural. Evaluations of prompt-level graph augmentations should report baseline accuracy, a length-matched control and a graph-blind solver.

Limitations

The main sweep covers two model sizes from one family; the screened arms extend the range but on a smaller corpus with fewer conditions and were not pre-registered. The route classification depends on a threshold and on the $n=40$ shortcut refit, and the route relation is not detectable within the 0.10–0.90 baseline window, where data are sparse. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the clustering primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches clustering to within 8% in tokens but is much shorter than rwse and all. Every prompt uses one encoding (incident), one template and one primer position, with greedy decoding. Several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample. Three further primers would address the main open questions: *shuffled-value* (a stricter placebo than filler), *corrupted* (to separate copying from computation) and *single-fact* (to separate interference from route substitution).

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

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A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.

Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

Arm	$p=0.1$		$p=0.2$		$p=0.3$		$p=0.5$		comp.	clust.	rwse	degree	filler	all
	F1	exact	F1	exact	F1	exact	F1	exact						
qwen3-1.7b	0.970	0.920	0.987	0.910	0.969	0.590	0.963	0.480	.076	1.00	1.00	1.00	1.00	1.00
qwen3-4b	0.985	0.930	0.986	0.880	0.985	0.840	0.993	0.790	.018	.148	.018	1.00	.018	1.00

Table 6: `connected_nodes` under `none`: set-F1, the task’s default metric, against exact match. F1 stays at 0.96–0.99 everywhere while exact match falls by 44 points for qwen3-1.7b, so the task looks ceilinged under F1 and is not.

(a) length cost (filler–none) by task			
Task	cells	mean	range
edge_existence	11	−9.1	[−32.0, −1.0]
connected_nodes	7	−6.3	[−14.0, +3.1]
node_degree	11	−4.3	[−17.2, +4.0]
node_count	1	+31.0	[+31.0, +31.0]

(b) content gain (cond. – filler) by arm			
Cond.	1.7b / 1.7b-think / 4b		
components	+5.7 / +6.8 / +5.5		
clustering	+10.9 / +2.6 / +11.6		
rwse	+11.3 / +5.6 / +5.2		

Table 7: Mid-range cells. (a) The cost of added text is task-specific, not a per-token rate, and changes sign within tasks, so it cannot be transferred across them. `node_count` rests on a single cell and is a constant-gold task; it is shown for completeness only. (b) Only `components` holds its content gain across arms; the ranking of the other two reverses between 1.7b and 4b, so no single ordering of primers by length survives disaggregation.

A.2 Robustness analyses

Arm	node_count	node_degree	edge_count	connected_nodes	cycle_check	all	cap
gemma4-12b	1.00	1.00	0.93	1.00	1.00	0.987	0/150
gemma4-12b-think	1.00	1.00	1.00	1.00	1.00	1.000	43/150
gemma4-e4b	1.00	1.00	0.96	1.00	0.97	0.986	5/150
gemma4-e4b-think	0.97	1.00	0.93	1.00	0.97	0.973	0/150
qwen3-8b	1.00	1.00	0.43	1.00	0.97	0.880	0/150
qwen3-8b-think	1.00	1.00	1.00	1.00	1.00	1.000	7/150
qwen3-14b	0.97	1.00	0.40	1.00	0.97	0.867	0/150
qwen3-14b-think	1.00	1.00	1.00	0.97	1.00	0.993	2/150

Table 5: The published GraphQA split (5–19 nodes, 30 graphs per cell) under `none`, exact match. Every arm scores 0.97–1.00 on four of the five tasks; only `edge_count` separates them, and only for the two plain Qwen arms (0.40–0.43 against 0.93–1.00 for the rest). `cap` counts generations that hit the token budget, excluded from the accuracies. This corpus cannot measure a primer effect, which is why the main sweep moves to $n=40$.

Table 8: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the screened arms. At $n=40$ (Table 1) the `edge_existence/none` bar rises to 0.735 through class imbalance, and `degree`’s gap over `none` on that task falls from +0.296 to +0.011.

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 9: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold**: above 50%.

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 10: `edge_existence`, pooled over the four main-sweep densities: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

Cond.	qwen3-1.7b			qwen3-4b		
	$\Delta d'$	Δc		$\Delta d'$	Δc	
components	−0.36	[−0.56, −0.21]	−0.18	+0.07	[−0.24, +0.33]	−0.10
clustering	+0.21	[+0.03, +0.41]	+0.11	+0.43	[+0.12, +0.75]	+0.08
rwse	+0.07	[−0.19, +0.43]	+0.28	−0.38	[−0.68, −0.15]	−0.32
degree	+0.15	[−0.05, +0.41]	+0.19	−0.04	[−0.33, +0.21]	−0.15
filler	−0.93	[−1.16, −0.71]	−0.46	−0.25	[−0.54, +0.00]	−0.25
all	+0.55	[+0.35, +0.77]	+0.27	−0.36	[−0.71, −0.11]	−0.31

Table 11: `edge_existence`, main sweep: change in sensitivity d' and criterion c against `none`, averaged over the four densities (log-linear correction; 95% bootstrap CI for $\Delta d'$). With the hit rate at ceiling, d' and c are not separately identifiable.

Accuracy under none	substitute route		no route	
	cells	Δ	cells	Δ
0.00–0.25	24	+6.3	24	−0.7
0.25–0.50	6	+8.5	6	+2.0
0.50–0.75	6	+15.5	24	+0.4
0.75–0.95	11	+4.2	33	−3.9
0.95–1.00	34	−9.0	186	−0.4
$r(\Delta, \text{baseline})$		−0.417		−0.050

Table 12: Mean primer effect (percentage points), binned by none accuracy on the same (arm, task, density, condition) cell of the main sweep, using the $n=40$ route split (Table 1). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

Estimate	r_{route}	k	r_{none}	k	interaction [95% CI]
naive, pp	−0.35	117	−0.01	381	−12.5 [−19.5, −6.3]
cross-fit (one direction), pp	−0.34	116	+0.03	381	−12.1 [−17.9, −7.0]
cross-fit (both), pp	−0.31	232	+0.02	762	−11.0 [−17.1, −5.7]
cross-fit (both), logit	−0.40	232	+0.04	762	−1.35 [−1.93, −0.77]
baseline .10–.90, pp	−0.01	78	−0.07	187	+3.0 [−29.1, +41.1]
baseline .10–.90, logit	+0.02	78	−0.05	187	+0.22 [−1.36, +1.94]
screened, pp	−0.67	105	−0.19	93	−24.7 [−109, −3.8]

Table 13: Route $\times$ baseline relation, main sweep plus high-density extension (screened: published split and probe). “Both” counts each cell once per fold direction; k for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for screened arms.

Comparison	Δ	n	density	p
clust. vs. none (main)	+3.0	400	.10–.50	0.26
clust. vs. none	+1.9	2800	.10–.85	0.022
filler vs. none	−2.4	2800	.10–.85	0.0013
clust. vs. filler	+4.3	2800	.10–.85	2.4×10^{-8}
clust. vs. none	+7.5	400	.35	0.0089
clust. vs. none (new seed)	+4.2	1600	.10–.50	0.00055

Table 14: qwen3-1.7b, node_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 15.

p	clust.–none	filler–none	clust.–filler
.10	+0.3	+3.0*	−2.8*
.20	+3.3	+0.5	+2.8
.35	+7.5*	+1.0	+6.5*
.50	+4.3	−2.0	+6.3*
.65	−2.0	−9.5*	+7.5*
.75	+0.5	−6.3*	+6.8*
.85	−0.5	−3.5*	+3.0*

Table 15: qwen3-1.7b, node_degree, dedicated sweep, 400 graphs per density. *: McNemar $p < 0.05$ (uncorrected).

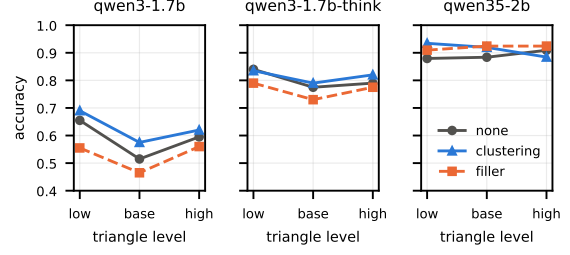


Figure 4: Rewiring: node_degree accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

Arm	low	base	high
predicted shift	−11.5	0.0	+16.7
qwen3-1.7b	+0.03	+0.28	−0.02
qwen3-1.7b-think	+0.01	+0.04	+0.01
qwen35-2b	−0.29	−0.26	−0.16

Table 16: Density-prior test on the rewiring corpus: mean paired shift in signed node_degree error, clustering minus none (errors clipped at ± 20 , 200 graphs per level). “Predicted” is the shift if answers moved fully to $\bar{C}(n-1)$.

Task	1.7B, acc. under none (%)			4B, acc. under none (%)		
	plain	think	think@2048	plain	think	think@2048
connected_nodes	72.5	78.7	67.2	86.0	95.5	89.0
cycle_check	90.7	97.0	38.0	99.0	99.3	62.0
edge_count	1.3	15.8	0.0	2.0	20.0	0.0
edge_existence	70.5	99.0	97.8	90.5	99.8	99.8
node_count	31.5	91.0	86.0	90.2	99.5	94.8
node_degree	60.5	76.2	67.7	99.3	97.0	91.2

Table 17: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Table 2.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
1.7B	node_count	0.69	0.47	0.01	0.67	0.62	0.29	0.22
	edge_count	133.4	98.1	105.7	79.0	47.7	181.2	94.2
	node_degree	0.94	0.91	0.78	0.91	0.67	1.27	0.91
4B	node_count	0.10	0.01	0.01	0.01	0.02	0.01	0.00
	edge_count	39.2	40.1	43.4	56.0	40.4	29.3	28.1
	node_degree	0.02	0.01	0.02	0.05	0.14	0.02	0.49

Table 18: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

A.3 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, a 95% paired-bootstrap interval, the permutation p -value, and Δ under the alternative convention that scores

truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[-2.8, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[-3.0, +5.8]	0.6458	+1.2
	rwse	400	+1.0	[-3.5, +5.2]	0.7304	+1.0
	degree	400	+3.5	[-0.5, +7.5]	0.1239	+3.5
	filler	400	-6.2	[-10.2, -2.2]	0.0040	-6.2
	all	399	+2.0	[-2.8, +6.8]	0.4644	+2.0
cycle_check	components	343	-0.6	[-1.5, +0.0]	0.5033	+2.0
	clustering	362	+0.3	[+0.0, +0.8]	1.0000	+8.5
	rwse	358	-7.5	[-10.3, -4.7]	0.0001	-0.2
	degree	301	-12.6	[-16.6, -9.0]	0.0001	-17.5
	filler	341	+0.3	[+0.0, +0.9]	1.0000	+3.0
	all	362	-1.1	[-2.5, +0.0]	0.2210	+7.0
edge_count	components	211	-1.4	[-3.8, +0.5]	0.3783	-1.0
	clustering	238	-0.4	[-2.5, +1.7]	1.0000	-0.5
	rwse	239	-0.8	[-2.9, +1.3]	0.6837	-0.8
	degree	253	+4.0	[+0.8, +7.1]	0.0179	+3.0
	filler	224	-0.4	[-2.2, +1.3]	1.0000	-0.5
	all	211	+2.4	[-0.5, +5.7]	0.2269	+0.8
edge_existence	components	400	-8.2	[-11.8, -4.8]	0.0001	-8.2
	clustering	400	+3.8	[-0.5, +8.0]	0.1199	+3.8
	rwse	399	+5.3	[+0.8, +9.8]	0.0321	+5.0
	degree	400	+4.0	[-0.5, +8.2]	0.0957	+4.0
	filler	400	-15.8	[-20.0, -11.8]	0.0001	-15.8
	all	400	+11.5	[+6.8, +16.2]	0.0001	+11.5
node_count	components	400	+21.2	[+15.8, +26.8]	0.0001	+21.2
	clustering	400	+67.2	[+62.7, +71.8]	0.0001	+67.2
	rwse	400	+2.0	[-4.2, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+3.0, +10.5]	0.0008	+6.8
	filler	400	+39.8	[+35.0, +44.8]	0.0001	+39.8
	all	400	+47.0	[+42.0, +52.0]	0.0001	+47.0
node_degree	components	400	+1.8	[-2.0, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[-1.8, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[-0.5, +8.5]	0.1104	+4.0
	degree	400	+7.5	[+3.0, +12.0]	0.0018	+7.5
	filler	400	-0.5	[-4.5, +3.5]	0.9028	-0.5
	all	400	+10.2	[+5.2, +15.2]	0.0003	+10.2

Table 19: qwen3-1.7b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[-0.5, +6.1]	0.1268	+2.5
	clustering	394	+1.5	[-2.5, +5.6]	0.5384	+2.0
	rwse	393	-0.3	[-4.1, +3.6]	1.0000	+0.2
	degree	390	+7.4	[+3.6, +11.3]	0.0001	+6.2
	filler	394	+2.0	[-1.5, +5.6]	0.3275	+2.5
	all	392	+6.1	[+2.6, +9.9]	0.0027	+6.2
cycle_check	components	346	+0.0	[+0.0, +0.0]	1.0000	-7.5
	clustering	376	+0.0	[+0.0, +0.0]	1.0000	-0.5
	rwse	386	+0.0	[+0.0, +0.0]	1.0000	+2.2
	degree	295	-0.7	[-1.7, +0.0]	0.5036	-21.8
	filler	388	+0.0	[+0.0, +0.0]	1.0000	+3.0
	all	373	+0.0	[+0.0, +0.0]	1.0000	-0.8
edge_count	components	27	+7.4	[+0.0, +18.5]	0.5037	-0.8
	clustering	23	-8.7	[-26.1, +8.7]	0.6251	-1.2
	rwse	25	-4.0	[-12.0, +0.0]	1.0000	-3.2
	degree	13	-61.5	[-84.6, -38.5]	0.0102	-3.2
	filler	29	+3.4	[-6.9, +13.8]	1.0000	-1.5
	all	24	-8.3	[-29.2, +12.5]	0.6794	+5.0
edge_existence	components	398	-0.8	[-2.3, +0.8]	0.5090	-0.8
	clustering	398	-1.0	[-2.5, +0.3]	0.2911	-1.0
	rwse	398	-1.3	[-3.0, +0.3]	0.2274	-1.2
	degree	397	-2.3	[-4.3, -0.5]	0.0249	-2.5
	filler	399	-0.8	[-2.3, +0.8]	0.5051	-0.5
	all	399	-2.0	[-3.8, -0.5]	0.0374	-1.8
node_count	components	342	+0.9	[-0.3, +2.3]	0.3758	+1.5
	clustering	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	rwse	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	degree	360	+0.8	[-0.6, +2.2]	0.4512	+5.5
	filler	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	all	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
node_degree	components	395	-1.0	[-4.3, +2.3]	0.6543	-1.0
	clustering	396	+1.0	[-2.8, +4.8]	0.7010	+1.5
	rwse	394	+2.5	[-1.5, +6.9]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +16.0]	0.0001	+8.8
	filler	396	-3.5	[-7.3, +0.3]	0.0954	-3.0
	all	396	+11.6	[+7.3, +15.9]	0.0001	+12.2

Table 20: qwen3-1.7b-think: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.0, +12.8]	0.0001	+9.2
	clustering	400	+1.0	[-3.0, +4.8]	0.6893	+1.0
	rwse	400	+0.0	[-4.0, +4.0]	1.0000	+0.0
	degree	400	-3.0	[-7.2, +1.2]	0.1882	-3.0
	filler	400	-12.5	[-17.0, -8.0]	0.0001	-12.5
	all	400	-2.8	[-7.0, +1.5]	0.2511	-2.8
cycle_check	components	391	-1.3	[-2.6, +0.0]	0.1318	-2.0
	clustering	395	+0.3	[+0.0, +0.8]	1.0000	+0.5
	rwse	392	+0.3	[+0.0, +0.8]	1.0000	-0.2
	degree	391	-1.5	[-2.8, -0.5]	0.0312	-2.2
	filler	392	-0.3	[-0.8, +0.0]	1.0000	-0.5
	all	395	-0.5	[-1.5, +0.5]	0.6269	-0.2
edge_count	components	397	-1.3	[-2.8, +0.3]	0.1756	-1.2
	clustering	399	-1.3	[-3.0, +0.3]	0.2298	-1.2
	rwse	398	-1.5	[-3.0, -0.3]	0.0668	-1.5
	degree	393	+27.7	[+23.2, +32.6]	0.0001	+27.3
	filler	400	+1.2	[-0.5, +3.2]	0.3012	+1.2
	all	397	+3.8	[+1.3, +6.3]	0.0049	+3.8
edge_existence	components	400	-2.8	[-6.0, +0.5]	0.1254	-2.8
	clustering	400	+4.0	[+1.0, +7.0]	0.0168	+4.0
	rwse	400	-10.0	[-14.0, -6.2]	0.0001	-10.0
	degree	398	-4.3	[-7.8, -1.0]	0.0200	-4.8
	filler	400	-7.0	[-10.8, -3.5]	0.0002	-7.0
	all	399	-9.8	[-13.8, -6.0]	0.0001	-10.0
node_count	components	400	+9.0	[+6.2, +12.0]	0.0001	+9.0
	clustering	400	+8.8	[+5.8, +11.8]	0.0001	+8.8
	rwse	400	+9.2	[+6.2, +12.2]	0.0001	+9.2
	degree	400	+8.0	[+5.0, +11.2]	0.0001	+8.0
	filler	400	+9.2	[+6.5, +12.2]	0.0001	+9.2
	all	400	+9.8	[+7.0, +12.8]	0.0001	+9.8
node_degree	components	400	+0.2	[+0.0, +0.8]	1.0000	+0.2
	clustering	400	-0.5	[-2.0, +0.8]	0.7246	-0.5
	rwse	400	-2.2	[-4.0, -0.5]	0.0240	-2.2
	degree	400	-6.5	[-9.2, -3.8]	0.0001	-6.5
	filler	400	-0.5	[-2.0, +0.8]	0.7232	-0.5
	all	400	-8.0	[-10.8, -5.2]	0.0001	-8.0

Table 21: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	[-0.5, +1.6]	0.6249	+0.5
	clustering	373	+0.0	[-1.3, +1.3]	1.0000	-0.8
	rwse	379	+0.0	[-1.3, +1.6]	1.0000	+0.5
	degree	376	+0.0	[-1.6, +1.6]	1.0000	+0.5
	filler	379	+0.8	[-0.3, +2.1]	0.3787	+1.2
	all	375	-0.5	[-2.1, +0.8]	0.7306	-1.2
cycle_check	components	381	+0.0	[+0.0, +0.0]	1.0000	-3.2
	clustering	394	+0.0	[+0.0, +0.0]	1.0000	+0.0
	rwse	395	+0.0	[+0.0, +0.0]	1.0000	+0.2
	degree	375	-0.8	[-1.9, +0.0]	0.2514	-5.5
	filler	396	+0.0	[+0.0, +0.0]	1.0000	+0.5
	all	392	+0.0	[+0.0, +0.0]	1.0000	-0.5
edge_count	components	60	+1.7	[+0.0, +5.0]	1.0000	+24.8
	clustering	59	+1.7	[-3.4, +6.8]	1.0000	+24.8
	rwse	49	+0.0	[-6.1, +6.1]	1.0000	+10.2
	degree	34	-23.5	[-38.2, -8.8]	0.0073	+8.5
	filler	50	-2.0	[-6.0, +0.0]	1.0000	+12.2
	all	41	-41.5	[-56.1, -26.8]	0.0002	+1.8
edge_existence	components	396	+0.0	[+0.0, +0.0]	1.0000	-0.5
	clustering	396	+0.0	[+0.0, +0.0]	1.0000	-0.8
	rwse	393	+0.0	[+0.0, +0.0]	1.0000	-1.2
	degree	398	+0.0	[+0.0, +0.0]	1.0000	+0.0
	filler	399	+0.0	[+0.0, +0.0]	1.0000	+0.0
	all	394	+0.0	[+0.0, +0.0]	1.0000	-1.0
node_count	components	400	+0.0	[-1.0, +1.0]	1.0000	+0.0
	clustering	399	+0.5	[+0.0, +1.3]	0.4936	+0.2
	rwse	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	degree	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	filler	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
	all	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
node_degree	components	380	-0.3	[-0.8, +0.0]	1.0000	+0.2
	clustering	386	-0.3	[-1.0, +0.5]	1.0000	+1.8
	rwse	381	-2.1	[-3.7, -0.8]	0.0072	-1.2
	degree	388	+0.0	[-0.8, +0.8]	1.0000	+2.5
	filler	386	+0.0	[+0.0, +0.0]	1.0000	+2.0
	all	383	-0.8	[-1.8, +0.0]	0.2466	+0.5

Table 22: qwen3-4b-think: all cells.